Hwee Jean (Cindy) Lim

EDUC 5030: Designing Effective Learning Environments

Project 1: Design Critique Report

Generative AI for Everyone by DeepLearning.AI

This critique examines *Week 1: Introduction to Generative AI*, the first module from the online course *Generative AI for Everyone* developed by DeepLearning.AI and delivered through the Coursera platform. The purpose of this report is to evaluate the instructional effectiveness of the unit using instructional design principles and frameworks introduced in this course, with attention to both strengths and areas for improvement.

The course is a three-week, self-paced massive open online course (MOOC) intended for a general audience with little to no prior technical background in artificial intelligence. Week 1 serves as the foundational unit and introduces core concepts related to generative AI, including large language models (LLMs), common use cases, limitations of current tools, and basic prompting strategies. According to the course description and learning objectives, this unit is designed to support primarily cognitive learning outcomes, with an emphasis on conceptual understanding rather than skill mastery or performance-based application.

DeepLearning.AI is the instructional design client, as the organization is responsible for the course's learning objectives, content structure, assessments, and instructional approach. Coursera is the delivery platform, particularly in relation to navigation, interface design, and learner experience constraints that may shape how the instruction is presented but are not fully controlled by the course designers. The scope of this critique is limited to Week 1 of the course, as this is the only unit accessible without paid enrollment. While the broader course structure is acknowledged when relevant, all evaluative claims and recommendations are grounded in observable elements from this initial module.

Instructional Value

Week 1 of *Generative AI for Everyone* appears to be designed to do more than simply introduce terminology. The unit aims to reshape how learners think about generative AI by

offering new mental models for understanding what large language models are, what they are good at, and where their limitations lie. The intended learning change is primarily cognitive, focusing on conceptual understanding rather than procedural skill or performance-based outcomes.

The learning objectives emphasize definition, recognition, and conceptual distinction, such as identifying generative AI tools, understanding how LLMs generate output, and recognizing common misconceptions. This level of cognitive demand is appropriate given the course's positioning as an entry-level MOOC for a broad audience. The unit prioritizes building a shared conceptual foundation, which is especially important in a domain where learners arrive with highly uneven prior knowledge.

The sequencing of content supports this goal. Early lessons focus on broad framing and simple explanations, such as describing LLMs as systems trained to predict the next word, before moving into more nuanced discussions of use cases, hallucinations, and prompting strategies. This progression allows learners to form an initial mental model and then gradually refine it. Short video segments further support comprehension by breaking content into manageable pieces, which aligns with cognitive load principles in asynchronous learning environments.

Assessments play a significant role in reinforcing instructional value. The quizzes are closely aligned with the lecture content and require learners to distinguish between plausible but incorrect statements and conceptually accurate ones. Questions addressing hallucinations, knowledge cutoffs, and general purpose technologies encourage learners to think critically rather than simply recall facts.

Appropriateness of Materials

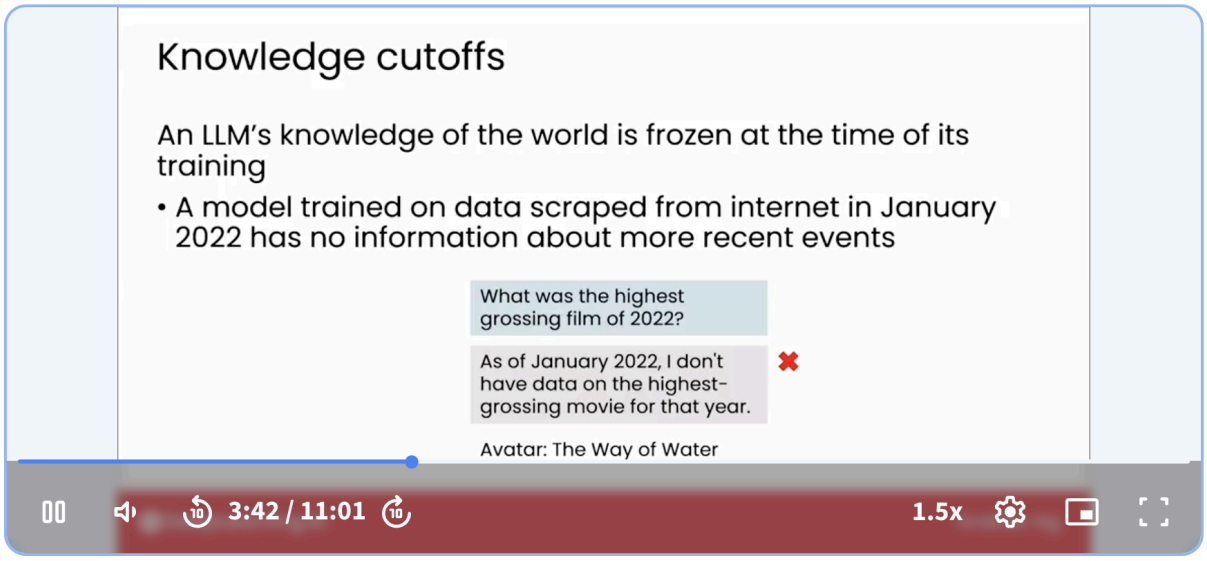
Much of Week 1's effectiveness comes from how deliberately the materials are scoped for a non-technical audience. The unit relies almost entirely on short lecture videos paired with concept-check quizzes, which keeps the instructional experience focused and predictable. There are no extraneous tools, simulations, or complex tasks introduced at this stage, which aligns with the course's stated goal of conceptual understanding rather than skill mastery.

In the opening videos, the instructor frames generative AI in everyday terms before introducing mechanisms or terminology. Rather than beginning with definitions, the course situates AI within familiar contexts such as writing, reading, and chatting tasks. This framing helps learners understand why generative AI matters before they are asked to understand how it works. The explanation of how generative AI works continues this pattern. Describing large language models as systems trained to repeatedly predict the next word offers a simplified but accurate mental model that appears consistently throughout the unit. This explanation avoids unnecessary technical detail while still supporting conceptual accuracy, which is appropriate for learners encountering these ideas for the first time.

In concept-check quizzes, learners are asked to identify accurate definitions of generative AI, recognize common misconceptions about LLM behavior, and distinguish between different types of AI applications. These questions closely mirror the language used in the lectures, which supports alignment between instruction and assessment. On the other hand, several quiz items emphasize limitations such as static knowledge cutoffs or lack of web access, which are illustrated in the lesson on what LLMs can and cannot do through examples of frozen training data and outdated responses (see Figure 1). While these limitations are conceptually important, they may not fully align with learners' experiences using newer AI systems. The course mitigates this tension through the "fresh college graduate" mental model, which reframes limitations as constraints of context and instruction rather than fixed tool deficiencies (see Figure 2), even as some examples may age more quickly than the concepts they are intended to illustrate.

Figure 1

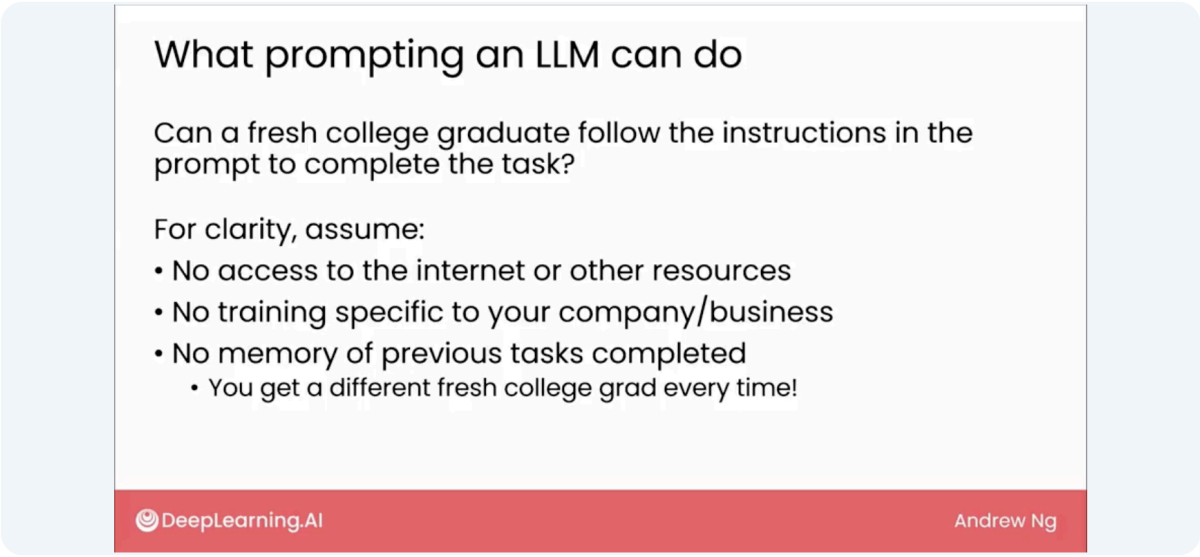
Course slide illustrating the concept of LLM knowledge cutoffs



What LLMs can and cannot do Save note

Figure 2

Slide introducing the "fresh college graduate" mental model for understanding LLM capabilities



What LLMs can and cannot do Save note

Ease of Use

Navigating Week 1 is straightforward, even for learners unfamiliar with Coursera or MOOCs. The unit follows a linear structure, with videos, quizzes, and optional resources clearly labeled and visually distinct. Learners can easily see how much content remains, which items are graded, and what is optional, supporting self-paced planning. The instructional materials themselves are also easy to process. The instructor uses consistent language across videos, returning repeatedly to the same explanations and metaphors rather than introducing new terminology for similar ideas. For learners new to AI, this repetition functions as reinforcement rather than redundancy.

One particularly effective design choice is the alignment between lecture language and quiz wording. Quiz questions often reuse phrasing introduced in the videos, reducing unnecessary cognitive load and helping learners focus on conceptual understanding rather than decoding unfamiliar language. Visual consistency further supports its usability. Slides, diagrams, and interface elements follow a uniform style, and there are no abrupt shifts in tone or presentation across lessons. This consistency makes the unit feel cohesive even as it moves across different topics, such as applications, limitations, and prompting strategies.

Some usability issues stem less from navigation and more from conceptual expectations. Learners who are already familiar with current AI tools may notice moments where the course's descriptions do not fully align with their lived experience. When quiz questions present limitations without contextual qualifiers, learners may briefly question whether the material is outdated or incorrect. While this does not prevent completion, it can interrupt the sense of coherence between instruction and reality.

Motivational Design

The course makes a clear effort to lower intimidation and anxiety around AI from the outset. Rather than positioning generative AI as a technical or elite domain, the instructor frames it as a general-purpose technology that many people are already encountering in everyday contexts. This framing helps learners feel included rather than excluded by the subject matter. The instructor presence also plays a significant role in sustaining engagement. The tone across videos is calm, conversational, and deliberately non-authoritative. The repeated use of the "fresh

college graduate" mental model gives learners a concrete way to reason about LLM capabilities without needing to understand the underlying mathematics or architecture.

The pacing of the unit also supports motivation. Short videos are interspersed with quizzes, which encourages active engagement and gives learners frequent opportunities to check their understanding. Opportunities for learner agency are present but underdeveloped. The course frequently references tools such as ChatGPT, Bard, or Bing Chat, and encourages learners to think about how they might use them. These suggestions remain largely implicit. More explicit prompts for optional experimentation or reflection could help learners connect the material to their own contexts without increasing instructional complexity.

Recommendations and Conclusion

Week 1 of *Generative AI for Everyone* is well designed for its intended audience and succeeds in establishing clear conceptual foundations through aligned objectives, accessible materials, and consistent instructional framing. The most significant challenge identified in this critique relates to the pace of change in generative AI rather than instructional misalignment. Explicitly distinguishing between enduring concepts and time-bound examples could help learners reconcile course content with their experiences using newer tools. Small design adjustments, such as annotations, update notes, or optional reflective prompts, could strengthen clarity and motivation while preserving the course's approachable tone. Overall, the unit demonstrates thoughtful instructional design choices that are well suited to an introductory MOOC context and could remain effective with modest, targeted refinements.

References

Generative AI for Everyone - Home - Week 1 | Coursera. (2021). Coursera. https://www.coursera.org/learn/generative-ai-for-everyone/home/module/1